

Reviewer Pack — Minimal Rank of Configuration Spaces vs Resolution Proof Len...

Entry 7b059428b180 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Configuration Spaces vs Resolution Proof Length for Tseitin Formulas

Entry ID: 7b059428b180 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 11:41:19 UTC

1. Conjecture

Field A (mathematical branch): Configuration Space Theory **Field B** (complexity object): Resolution Proof Complexity for Tseitin Formulas

Statement:

[‘For any instance G of Tseitin formulas, the resolution proof length is lower-bounded by $2^{(\min \text{Rank}(\text{config_space}(G)))}$ ’, “where ‘ $\text{Rank}(\text{config_space}(G))$ ’ denotes the minimal rank of the configuration space of G .”, ‘Equivalently, there exists a polynomial-time computable function f such that for all instances G , resolution proof length $\geq 2^{f(\min \text{Rank}(\text{config_space}(G)))}$.’]

Rationale (proposer’s reasoning):

[‘Configuration spaces encode geometric and topological properties of combinatorial objects, and minimal rank could capture the complexity inherent in the structure of Tseitin formulas.’, ‘By leveraging this connection, we may uncover structural insights that are not apparent through traditional complexity measures.’]

Taxonomy category: TSEITIN_RES (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): c9a002d144705f64

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all instances G with $n \leq 40$, the ratio of the resolution proof length to $2^{(\min \text{Rank}(\text{config_space}(G)))}$ is greater than or equal to 0.8 AND the average of this ratio across all seeds is less than or equal to 3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "configuration space theory" AND "resolution proof complexity" AND Tseitin formulas" - "minimal rank configuration space" AND Tseitin formulas AND resolution proof" - "Tseitin formulas resolution proof length" AND lower-bound minimal rank configuration space

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_tseitin_formula(n):
        variables = list(range(1, n + 2))
        clauses = []
        for i in range(1, n + 1):
            clauses.append([variables[i - 1], variables[n + i - 1]])
            clauses.append([-variables[i - 1], -variables[n + i - 1]])
            for j in range(i + 1, n + 1):
                clauses.append([-variables[i - 1], variables[j - 1], -variables[n + j - 1]])
        return clauses

    def compute_min_rank(clauses):
        # Simplified homology computation (not actual rank)
        return len(clauses) // 2

    def resolution_length(clauses):
        stack = []
        for clause in clauses:
            if not any(abs(lit) in [abs(x) for x in stack] for lit in clause):
                stack.append(clause[0])
        return len(stack)

    n = random.randint(5, 40)
    formula = generate_tseitin_formula(n)
    min_rank = compute_min_rank(formula)
    proof_length = resolution_length(formula)
```

```

if min_rank == 0:
    return {
        "metric_name": "Ratio of Resolution Proof Length to 2^(Min Rank)",
        "metric_value": None,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "min_rank is zero"
    }

ratio = proof_length / (2 ** min_rank)
return {
    "metric_name": "Ratio of Resolution Proof Length to 2^(Min Rank)",
    "metric_value": ratio,
    "instances_tested": 1,
    "conjecture_holds": ratio >= 0.8,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all(r["metric_value"] is not None and r["conjecture_holds"] for r in results):
        mean_ratio = sum(r["metric_value"] for r in results) / len(results)
        support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)
        print(f"RESULT: SUPPORTED mean={mean_ratio} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not (r["metric_value"] is not None and r["conjecture_holds"])))
        print(f"RESULT: FALSIFIED counterexample='min_rank_zero' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_63ad6674.py", line 71, in <module>

```

    result = run_trial(seed)
    ~~~~~

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_63ad6674.py", line 43, in run_trial

```

    formula = generate_tseitin_formula(n)
    ~~~~~

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_63ad6674.py", line 28, in generate_tseitin

```

    clauses.append([-variables[i - 1], variables[j - 1], -variables[n + j - 1]])
    ~~~~~

```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means that the pre-registered support condition could not be unambiguously met. | next: Re-run the test with a different seed or investigate the cause of the crash to ensure that it does not affect the validity of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14491
2	propose	ollama_remote	glm4:latest	0	0	13073
3	propose	ollama_remote	glm4:latest	0	0	13423
4	preregistration	ollama_remote	glm4:latest	0	0	10333
5	novelty	ollama_remote	glm4:latest	0	0	8359
6	novelty	ollama_remote	glm4:latest	0	0	8924
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14554
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11265
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9774
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8219
11	judge	ollama_remote	glm4:latest	0	0	12006

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 124419 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/7b059428b180.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/7b059428b180.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/7b059428b180.tar.gz` (if generated)